

Dr. Romain Mottier

Computational Scientist — PhD in Applied Mathematics

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French (native) • English (C2) • French citizen – Swiss work permit (G)

PROFILE

Computational Scientist in Applied Mathematics & **Physics.** Five years developing and shipping simulation code for engineering and physics problems: wave propagation, solid–fluid interaction, geophysics and electromagnetics. Production-level **C++ and Fortran**, Python for scripting and post-processing, from **PDE solver design** and mathematical validation to large-scale parallel runs (**MPI/OpenMP**) on HPC clusters. Contributor to **SEM3D**, a production seismic simulation code used for nuclear-site studies.

Position sought. R&D and scientific engineering roles where rigour translates into industrial applications. Comfortable entering a new physical or mathematical domain from scratch: four distinct fields in five years, each carried through to working, validated code.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher — Computational Mathematics 10/2025 – present
University of Basel, Dept. of Mathematics and Computer Science Basel, Switzerland

- Extending **DiSk++** (C++17 finite element library) with multilevel explicit local time-stepping, so a locally refined mesh no longer forces the whole domain down to the smallest time step.
- Developing high-order solvers for strongly nonlinear PDEs whose nonlinearity varies in space.
- Work presented at **three international conferences** in 2026 (Delft, Vienna, Montreal); two research articles in preparation.
- Sole lecturer for a final-year BSc course in numerical analysis of ODEs (Fall 2026), designed from scratch — lectures, exercises and assessment.
- Supervised **about ten Master's students** on semester projects in numerical simulation (finite elements for linear elasticity; spectral-element solver for acoustic waves), from project definition to final report.

PhD Researcher — Applied Mathematics & Scientific Computing 10/2021 – 07/2025
French Atomic Energy Commission (CEA) & École nationale des Ponts et Chaussées (ENPC) & Polytechnic Institute of Paris (IP Paris) Paris, France

- Developed and analysed high-order finite element methods for coupled **elasto-acoustic (solid–fluid) wave propagation**, carried from mathematical analysis through to validated implementations in two codes (C++ and Fortran).
- Ran 3D seismic wave simulations on the **CEA EXA1 supercomputer**: tens of millions of unknowns on 1,000+ cores (MPI/OpenMP).
- Worked across three institutions, reporting to both academic supervisors and CEA researchers.
- Authored **3 research articles** published in leading international journals and presented at **5 international and national conferences**.
- Taught tutorials and **Python** lab sessions in numerical methods at Paris Dauphine and Sorbonne University, at BSc and MSc level.

Research Intern — Computational Electromagnetics 03/2021 – 08/2021
ONERA, The French Aerospace Lab Toulouse, France

- Implemented Compatible Discrete Operator (CDO) schemes for the time-domain Maxwell equations, benchmarked against reference solutions (Fortran 90).

Research Intern — Planetary Thermal Modelling 05/2020 – 08/2020
European Space Agency (ESA/ESTEC) Noordwijk, Netherlands

- Implemented a numerical heat-transfer model of the surface and sub-surface temperature distribution on Mercury (Fortran 90).

SOFTWARE PROJECTS

DiSk++ github.com/wareHHOuse/diskpp

C++17 template library for discontinuous skeletal methods (HHO/HDG)

- Contributed the **elasto-acoustic wave propagation module** to an open-source scientific library. Implemented templated C++ covering assembly, static condensation and explicit/implicit time integration, dimension- and mesh-agnostic by design — supporting the results published in **M2AN** (2025) and **CMAME** (2026).
- Built on Eigen, MUMPS, SILO/HDF5, Gmsh and CMake; validated against analytical solutions on 2D/3D benchmarks up to order 7, with problems of several million degrees of freedom.

SEM3D github.com/sem3d/SEM

Fortran 90 spectral-element solver for 3D seismic wave propagation

- Implemented the HDG method for the elastodynamic wave equation, with explicit time integration and MPI/OpenMP parallelisation, in a **production geophysics code** developed by CEA and academic partners for broadband (0–10 Hz) regional-scale seismic simulation at nuclear sites.

ProtoN github.com/wareHHOuse/ProtoN

C++ library for prototyping cut/unfitted finite element methods

- Implemented discontinuous cut-FEM (unfitted HHO) schemes with polynomial-extension stabilisation for elliptic interface problems, supporting the results published in **SINUM** (2026).

TECHNICAL SKILLS

Programming	**C++** • **Python** • **Fortran** • MATLAB • Bash
Applied mathematics	Partial Differential Equations • Numerical Analysis • Linear Algebra • Optimisation
Scientific computing	High-Performance Computing (MPI, OpenMP) • Eigen • Intel MKL • MUMPS • Gmsh
Numerical methods	Finite Element & Finite Volume Methods • Time Integration • Sparse Linear Solvers
Data & analysis	NumPy • Pandas • Matplotlib • ParaView
Software engineering	Git • CMake • Unit testing • Code review • Linux • LLM coding assistants (Claude Code, Copilot)
Application domains	Wave Propagation & Solid–Fluid Interaction • Geophysics • Electromagnetics

EDUCATION

PhD in Applied Mathematics & Scientific Computing 10/2021 – 07/2025

French Atomic Energy Commission (CEA) & École nationale des Ponts et Chaussées (ENPC) & Polytechnic Institute of Paris (IP Paris) Paris, France

- Thesis: *Hybrid high-order methods for the numerical simulation of elasto-acoustic wave propagation* Defended July, 2025, supervised by A. Ern and L. Guillot.

Engineering Diploma (MSc equivalent) — Modelling and Fluid–Structure Computation 09/2018 – 09/2021

Université de Toulon, SeaTech Engineering School Toulon, France

- Exchange semester — *MSc Numerical Methods in Engineering*, Universitat Politècnica de Catalunya (UPC), Barcelona, Spain (09/2020 – 02/2021).

PUBLICATIONS & CONFERENCES

1. Unfitted hybrid high-order methods stabilized by polynomial extension for elliptic interface problems. *With E. Burman and A. Ern* — published in **SIAM Journal on Numerical Analysis**, 2026
2. Hybrid high-order methods for elasto-acoustic wave propagation in the time domain. *With A. Ern, R. Khot and L. Guillot* — published in **ESAIM: Mathematical Modelling and Numerical Analysis**, 2025
3. Elasto-acoustic wave propagation in geophysical media using hybrid high-order methods on general meshes. *With A. Ern and L. Guillot* — published in **Computer Methods in Applied Mechanics and Engineering**, 2026
4. Multilevel explicit local time-stepping for wave propagation with high-order hybrid nonconforming schemes (HDG/HHO). *With M. J. Grote — in preparation*
5. A hybrid high-order method for variable-exponent Leray–Lions problems. *With A. Harnist — in preparation*

International & National Conference presentations

Talks — Wave-NED (Delft, 08/2026) • CMAM (Vienna, 07/2026) • WAVES (Montreal, 06/2026) • WCCM (Vancouver, 07/2024) • ECCOMAS (Lisbon, 06/2024) • CANUM (Île de Ré, 05/2024).

Posters — AGU Fall Meeting (San Francisco, 12/2023) • CJCMA (Paris, 09/2023).

REFERENCES

Contact details available on request

Prof. Marcus J. Grote — Postdoctoral advisor
Professor, University of Basel

Dr. Laurent Guillot — Industrial PhD advisor
Researcher, CEA

Prof. Alexandre Ern — PhD advisor
Senior Researcher, INRIA (SERENA)

Dr. Guillaume Delay
Researcher, Sorbonne University (LJLL)